

Group 14

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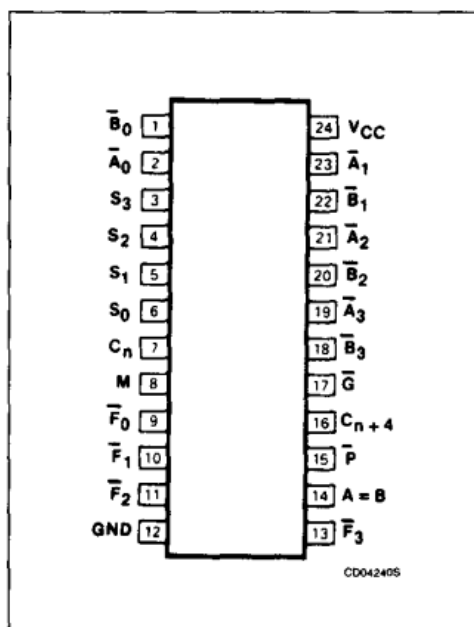
Hardware Lab Report

The IC 74181 is a 4-bit arithmetic logic unit (ALU) integrated circuit (IC) that was widely used in early microprocessor systems. The 74181 ALU is designed to perform various arithmetic and logic operations on 4-bit binary numbers.

Here are some key features and characteristics of the IC 74181:

- 4-Bit Inputs and Outputs:** The IC 74181 operates on 4-bit binary numbers, which means it can perform operations on four binary digits (bits) at a time. It has two 4-bit data inputs (A and B) and one 4-bit data output (F).
- Arithmetic Operations:** The 74181 can perform basic arithmetic operations such as addition, subtraction, and logic operations like AND, OR, XOR.
- Function Selection:** It has several control inputs to select the specific operation to be performed. These control inputs include M (mode control), S (subtraction), P (propagate), and G (generate). By setting these inputs to specific values, you can choose the desired operation.
- Carry and Overflow Outputs:** The IC provides a carry output (C_n) for carry propagation during addition and an overflow output (V) to indicate overflow conditions.

PIN CONFIGURATION



FEATURES

- Provides 16 arithmetic operations: ADD, SUBTRACT, COMPARE, DOUBLE, plus 12 other arithmetic operations
- Provides all 16 logic operations of two variables: Exclusive-OR, Compare, AND, NAND, NOR, OR, plus 10 other logic operations
- Full lookahead carry for high-speed arithmetic operation on long words

DESCRIPTION

The '181 is a 4-bit high-speed parallel Arithmetic Logic Unit (ALU). Controlled by the four Function Select inputs ($S_0 - S_3$) and the Mode Control Input (M), it can perform all the 16 possible logic operations or 16 different arithmetic operations on active HIGH or active LOW operands. The Function Table lists these operations.

MODE SELECT — FUNCTION TABLE

MODE SELECT INPUTS				ACTIVE HIGH INPUTS & OUTPUTS	
S ₃	S ₂	S ₁	S ₀	Logic (M = H)	Arithmetic** (M = L) (C _n = H)
L	L	L	L	$\bar{A}$	A
L	L	L	H	$\overline{A+B}$	A + B
L	L	H	L	$\overline{AB}$	A + $\bar{B}$
L	L	H	H	Logical 0	minus 1
L	H	L	L	$\overline{AB}$	A plus $A\bar{B}$
L	H	L	H	$\bar{B}$	(A + B) plus $A\bar{B}$
L	H	H	L	$A \bullet B$	A minus B minus 1
L	H	H	H	$A\bar{B}$	AB minus 1
H	L	L	L	$\overline{A+B}$	A plus AB
H	L	L	H	$\overline{A \bullet B}$	A plus B
H	L	H	L	B	(A + $\bar{B}$) plus AB
H	L	H	H	AB	AB minus 1
H	H	L	L	Logical 1	A plus A*
H	H	L	H	$A + \bar{B}$	(A + B) plus A
H	H	H	L	A + B	(A + $\bar{B}$) plus A
H	H	H	H	A	A minus 1

MODE SELECT INPUTS				ACTIVE LOW INPUTS & OUTPUTS	
S ₃	S ₂	S ₁	S ₀	Logic (M = H)	Arithmetic** (M = L) (C _n = L)
L	L	L	L	$\bar{A}$	A minus 1
L	L	L	H	$\overline{AB}$	AB minus 1
L	L	H	L	$\overline{A+B}$	$A\bar{B}$ minus 1
L	L	H	H	Logical 1	minus 1
L	H	L	L	$\overline{A+B}$	A plus (A + $\bar{B}$)
L	H	L	H	$\bar{B}$	AB plus (A + $\bar{B}$)
L	H	H	L	$\overline{A \bullet B}$	A minus B minus 1
L	H	H	H	$A + \bar{B}$	A + $\bar{B}$
H	L	L	L	$\overline{AB}$	A plus (A + B)
H	L	L	H	$\overline{A \bullet B}$	A plus B
H	L	H	L	B	$A\bar{B}$ (A + B)
H	L	H	H	A + B	A + B
H	H	L	L	Logical 0	A plus A*
H	H	L	H	$A\bar{B}$	AB plus A
H	H	H	L	AB	$A\bar{B}$ plus A
H	H	H	H	A	A

L = LOW voltage

H = HIGH voltage level

*Each bit is shifted to the next more significant position.

**Arithmetic operations expressed in 2s complement notation.

